

Cochain conventions for the independent space-group calculation

AllFSPT calculation note

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1 Physical layers and the equations being solved

In three spatial dimensions the decorated state is $(n_1, n_2, n_3, \hat{\nu}_4)$, where n_1 counts integer $p + ip$ sheets, n_2 is the Majorana-chain decoration, n_3 is the complex-fermion decoration, and $\hat{\nu}_4$ is an additive phase. Their coefficients are

$$n_1 \in C^1(G_b; \mathbb{Z}_{s_1}), \quad n_2 \in C^2(G_b; \mathbb{F}_2), \quad n_3 \in C^3(G_b; \mathbb{F}_2), \quad \hat{\nu}_4 \in C^4(G_b; \mathbb{R}/\mathbb{Z}_{s_1}). \quad (1)$$

The extension ω_2 and character s_1 are held fixed. The current space-group calculation uses $\omega_2 = 0$ and the binary orientation character defined by $(-1)^{s_1(g)} = \det(g)$. We display backgrounds in the formulas so that this specialization is visible. The lower equations are

$$d_{s_1} n_1 = 0 \quad \text{over } \mathbb{Z}, \quad (2)$$

$$dn_2 = \omega_2 \cup n_1 + s_1 \cup n_1^2 \quad \text{over } \mathbb{F}_2, \quad (3)$$

$$dn_3 = \mathcal{O}_4[n_1, n_2], \quad (4)$$

$$d_{s_1} \hat{\nu}_4 = \hat{\mathcal{O}}_5[n_1, n_2, n_3] \pmod{1}. \quad (5)$$

Binary occurrences of n_1 mean its parity; the integer itself is retained where a lift carry or fractional phase requires it.

2 Cochain arithmetic

The lift $\tilde{z}$ is taken after the entire expression z is formed over $\mathbb{F}_2$. Integer cup products use signed interval cuts. The differential on phases is $d_{s_1} = d - 2\tilde{s}_1 \cup$. For nonclosed cochains we use

$$S^j(u) = u \cup_{|u|-j} u + u \cup_{|u|-j+1} du, \quad dS^j(u) = \text{Sq}^j(du). \quad (6)$$

Negative cup indices give zero. The second term must not be omitted when the lower equation makes u nonclosed. Exact divisions precede reduction; negative carries use floor rather than truncation toward zero. The displayed obstruction and product formulas have denominators dividing eight or sixteen. Their solved phase primitives are general exact rational cochains modulo one, without this denominator restriction.

3 The closed Majorana sector

For a closed Majorana cochain a of degree $p = 1, 2$ and a complex-fermion cochain c of degree $p + 1$, set

$$P(a) = \text{Sq}^2 a + \omega_2 \cup a + s_1 \cup \text{Sq}^1 a, \quad dc = P(a). \quad (7)$$

The total phase in the Gu–Wen (CA) coordinate is

$$\widehat{O}_{p+3}(a, c) = \frac{1}{2}[\mathbf{S}^2(c) + \omega_2 \cup c] + \widehat{O}_{p+3}^\gamma(a). \quad (8)$$

The full Majorana term is the current manuscript Cartan–Adem expression. Its ordered words, Bockstein terms, and all antiunitary terms are evaluated directly by the independent expression graph in `fspt/formulas.py`. No microscopic phase table is queried.

For two states define

$$m(a, b) = a \cup_{p-1} b + s_1 \cup (a \cup_p b), \quad A = a + b, \quad C = c + c' + m(a, b). \quad (9)$$

The phase product is $\widehat{\nu} + \widehat{\nu}' + U_{p+2}(a, c; b, c')$. The complete U_3, U_4 are evaluated as the six manuscript blocks: Gu–Wen, Bockstein, Adem, extension, antiunitary, and mixed background. They satisfy

$$d_{s_1} U_{p+2} = \widehat{O}_{p+3}(A, C) - \widehat{O}_{p+3}(a, c) - \widehat{O}_{p+3}(b, c'). \quad (10)$$

This identity alone does not select a closed product correction. The reference dictionary is retained when matching the physical operator coordinate:

$$\Gamma(c) = \frac{1}{2}c \cup_{p+1} dc, \quad O_{\text{op}} = O_{\text{CA}} + d_{s_1}\Gamma, \quad U_{\text{op}} = U_{\text{CA}} + \Delta\Gamma. \quad (11)$$

The source comparison also includes its specified closed unary phase Z . It is not dropped because the obstruction is unchanged.

4 The native integer-layer secondary source

Put

$$h_1 = \llbracket n_1/2 \rrbracket_2, \quad u = n_2 + s_1 \cup h_1, \quad W = \omega_2 + s_1^2. \quad (12)$$

Then $du = W \cup n_1$, and the native short source is

$$\begin{aligned} \mathcal{O}_4[n_1, n_2] = & \mathbf{S}^2(u) + \omega_2 \cup u + s_1 \cup \mathbf{S}^1(u) + \zeta_{2,1}(W, n_1) \\ & + (W \cup_1 W + s_1 \cup W) \cup h_1 \\ & + [(W \cup_1 W) \cup_1 s_1 + s_1 \cup (s_1 \cup_1 W)] \cup n_1. \end{aligned} \quad (13)$$

The Cartan term is the literal operation $\zeta_{2,1}(W, a) = \smile_{123134} (W, W, a, a)$. This fixes a cochain representative, not merely its cohomology class. In particular the uniform higher-dimensional source is not substituted without its accompanying coboundary dictionary.

5 The complete integer-layer terminal phase

Write $A = du$ and define the integer quotients

$$K_3 = \frac{\widetilde{A} - \widetilde{W} \cup n_1}{2}, \quad \beta^\circ u = \frac{d\widetilde{u} - \widetilde{du}}{2}. \quad (14)$$

The second quotient is valid for nonclosed u ; $d\widetilde{u}/2$ alone is not. The terminal phase is assembled as

$$\widehat{O}_5 = \frac{1}{2}[\mathbf{S}^2(n_3) + \omega_2 \cup n_3] + \widehat{O}_5^{\gamma, \text{ext}}(u) + \widehat{O}_5^{\gamma\psi} + \widehat{O}_5^\psi. \quad (15)$$

Here the mixed term is

$$\widehat{\mathcal{O}}_5^{\gamma\psi} = \frac{1}{2}[\mathbf{S}^2 u + \omega_2 \cup u + s_1 \cup \mathbf{S}^1 u] \cup_3 \mathcal{O}_4^\psi[n_1]. \quad (16)$$

The pure integer term contains the fixed completion $y_5 = H_{\text{EZ}}^* \mathcal{R}_6 + AW^* y_5^{\text{tot}}$, the quadratic K_3 terms, the whole lifted Cartan bracket, and the sixteenth-valued coupling

$$\frac{1}{16}[\widetilde{W} \cup \widetilde{W} + \widetilde{W} \cup_1 d_{s_1} \widetilde{W}] \cup n_1. \quad (17)$$

All 55 fixed coefficients of y_5^{tot} and the complete individual terms are those in the supplied normalized note. They are compilation inputs, not unknown primitives to solve separately on each group.

The independent native evaluator compiles this entire formula into 7,239 scalar integer operations for $\omega_2 = 0$. The actual integer n_1 , both lower choices, transport signs, and carry bits are retained. The compiled evaluator has no dependency on the old SptSet engine or on a table of space-group classifications. For the canonical torsion representative, the exact restriction is

$$\widehat{O}_5(s_1, B, C) = \frac{1}{2}\mathbf{S}^2(C) + \widehat{O}_5^{\gamma, \text{ext}}(B) + \frac{9}{16}s_1^5. \quad (18)$$

Indeed $h_1 = 0$, $\mathcal{O}_4^\psi[s_1] = 0$, and the normalized pure phase pulls back from BC_2 as $9s_1^5/16$. Every local sign simplex verifies these restrictions against the unchanged supplied evaluator. A 935-instruction evaluator implements this case; 512 further local towers agree exactly with the generic expression.

6 From cochains to classification and stacking

The native free resolution is used for exact cocycles, primitives and quotient coordinates. A primitive obtained there is corrected by the comparison homotopy before it is used as a bar cochain. This correction ensures that the displayed lower equations hold on the simplices where the nonlinear formula is evaluated.

For the native pure-CF construction there is an additional comparison between the two chosen Steenrod cochain representatives. Let C be closed and let ν_0 be the bar image of the native phase seed, with the Majorana and extension inputs zero. The exact reconstruction solves

$$\rho_5 = [2(O_{\text{bar}}(0, C) - d_{s_1} \nu_0)]_2, \quad dh_4 = \rho_5, \quad \nu_4 = \nu_0 + \frac{1}{2}h_4 \pmod{1}. \quad (19)$$

Half-integrality and exact solvability are checked. The result satisfies $d_{s_1} \nu_4 = O_{\text{bar}}(0, C)$, while $2\nu_4 = 2\nu_0$ modulo one. Thus the comparison term cancels under doubling but is needed for the actual lift. Agreement of native and bar operations in cohomology alone does not identify their raw cochain vectors.

Changing the Majorana primitive changes the secondary source by the appropriate lower operation. The d_3 test therefore includes all relevant Majorana shifts; the d_4 test projects through the CF and Majorana-secondary images. A single nonzero raw phase vector is not a proof that the integer decoration is killed.

For the incoming gauge let $P = P(a_1)$. The state $(0, P, O_4(a_1, 0))$ has the required phase boundary. If $dc_2 = P$, composing it with the CF gauge differs from $O_4(a_1, c_2)$ by

$$\frac{1}{2}P \cup_2 P = d_{s_1} \left[\frac{1}{2}\mathbf{S}^1(c_2) \right]. \quad (20)$$

This exact witness aligns the group presentation with the chosen phase coordinate.

The open-Majorana continuation also has a nontrivial zero-integer coordinate dictionary. For $dB = 0$ and $dC = \text{Sq}^2 B + s \text{Sq}^1 B$, we obtain

$$O_5^{\text{pip}}(0, B, C) - O_5^{\text{CA}}(B, C) = d_{s_1} \left[D_0(s, B) + \frac{1}{4}\widetilde{\text{Sq}^2 B} + \frac{1}{2}s \cup C \right]. \quad (21)$$

The quarter-valued local cochain D_0 solves all 32,768 closed- B five-simplex equations. Another 128 complete legal towers verify the identity. Thus the two formulas define the same secondary cohomology class; their displayed phase cochains must not be identified pointwise.

7 The torsion integer-layer square

The supplied U_3, U_4 set the integer layer to zero. The collaborator's calibrated cochain product is an additional mathematical input; its universal coefficients are independently evaluated here. We align that reference product with the normalized formulas before using its diagonal on $n_1 = s_1$, with $\omega_2 = 0$. Write $s = s_1$, $B = n_2$, $C = n_3$. The CF coordinate dictionary is $C_{\text{ref}} = C + \Lambda$, where

$$\Lambda_{0123} = B_{013} + B_{012}B_{013} + B_{012}B_{023} + B_{013}B_{023}. \quad (22)$$

It satisfies $d\Lambda = Q_{\text{native}} + Q_{\text{ref}}$. The phase dictionary is

$$T_1(C) = D_4(s, B) + \frac{1}{4}s^4 + \frac{1}{2}[C \cup_2 \Lambda + \Lambda \cup_3 Q_{\text{native}}], \quad (23)$$

$$O_{\text{ref}}(s, B, C + \Lambda) - O_{\text{native}}(s, B, C) = d_{s_1}T_1(C). \quad (24)$$

The fixed local cochain D_4 is tabulated on four sign edges and six Majorana cone faces. Its 1,024 sixteenth-valued entries retain the initial AW coordinate; the explicit $s^4/4$ converts this base to the corrected supplied coordinate. Their sum solves all 32,768 legal local five-simplex equations. The additional C dependence follows from the polarization identity for S^2 . No target-group answer enters this universal cochain dictionary.

For the doubled output $n_1 = 2s, B = 0$, one has $\Lambda_2 = s^3$ and

$$T_2(C) = -\frac{3}{16}s^4 + \frac{1}{2}C \cup_2 s^3. \quad (25)$$

The restriction $B = 0$ is part of this formula's stated domain.

To remove $2s$, let t be the height zero-cochain on the cylinder and K its prism contraction to the bottom. The legal cylinder tower is

$$n_I = 2s + d_{s_1}t, \quad B_I = s \cup t \cup dt, \quad C_I = \pi^*C + KQ_{\text{native}}(n_I, B_I). \quad (26)$$

The endpoints are $(2s, 0, C)$ and $(0, 0, C + s^3)$. Its phase transport is $g_4 = \int_I O_5(n_I, B_I, C_I)$, satisfying

$$d_{s_1}g_4 = O_5(0, 0, C + s^3) - O_5(2s, 0, C). \quad (27)$$

The resulting $g_4 = 13s^4/16 + P(s, C)/2$ is an explicit binary polynomial in four sign edges and four CF cone faces. Its complete coefficients are recorded with the code; all 32,768 legal local five-simplex states verify the displayed transport identity.

If the reference diagonal product has CF output β and phase correction Γ_{ref} , its normalized lower-sector square is

$$(0, \beta, 2\widehat{\nu} + 2T_1(C) + \Gamma_{\text{ref}} - T_2(\beta + s^3) + g_4(\beta + s^3)). \quad (28)$$

This retains the reference product's closed phase correction. The extension order is subsequently obtained by reduction through the actual lower-layer generators; no order-two lift is presumed.

7.1 AW transport and phase convention

Each integer edge in the AW middle block is transported from its own initial vertex: $M_{ij} = (-1)^{s_{0i}} n_{ij}$. Correcting an earlier compiler transcription changes the phase, at $\omega_2 = 0$, by

$$\Delta O_5(012345) = \frac{1}{2} s_{01} s_{12} s_{23} s_{34} \text{bit}_1((-1)^{s_{03}} n_{34}) (n_{45} \bmod 2) \pmod{1}. \quad (29)$$

This is zero for every even n and for $s = 0$. At $n = s$ it is $s^5/2$, so the corrected flat phase is $\nu_{\text{new}} = \nu_{\text{old}} - s^4/4$. The unary dictionary changes by $T_{1,\text{new}} = T_{1,\text{old}} + s^4/4$; the canonical cylinder changes by $g_{\text{new}} = g_{\text{old}} + s^4/2$. The reference Γ_{ref} is unchanged. Therefore the complete square changes only by

$$\nu_{\text{square,new}} - \nu_{\text{square,old}} = \frac{1}{2} s^4 = d_{s_1} \left[\frac{1}{2} s \cup B \right], \quad dB = s^3. \quad (30)$$

This is an explicit phase gauge, proving equality of the marked relation classes after the input lifts are rephased. Saved earlier cochains are not current-coordinate witnesses. Direct comparison with the unchanged supplied evaluator now includes 84 signed towers, negative integers up to 2^{45} , all sign simplices, and both original failing regression fixtures.

8 Primitive free integer-layer generators

The free rank alone does not fix the lattice of surviving integer decorations. Put $H = H^1(G, \mathbb{Z}_s)$, and let $S \subset H$ be the subgroup surviving all outgoing differentials. For $s = 0$, every integral character has the literal flat tower

$$(n_1, n_2, n_3, \widehat{\nu}_4) = (n_1, 0, 0, 0). \quad (31)$$

In particular, substituting $s = n_2 = n_3 = 0$ into the complete normalized O_5 expression gives zero with n_1 left arbitrary. The native integral cohomology basis therefore gives primitive free generators in this case.

For a signed character, the cocycle equation defines a homomorphism

$$G \longrightarrow D_\infty = \mathbb{Z} \rtimes C_2, \quad g \longmapsto (n_1(g), s(g)). \quad (32)$$

Let x denote the universal signed integer coordinate on D_∞ . The class $2x$ has zero first source, so $n_2 = 0$ is allowed. Since $D_\infty = C_2 * C_2$, restriction to the two reflection groups is an isomorphism in cohomological degrees at least two. On these subgroups, $2x$ restricts respectively to 0 and $2s$, and the normalized parity source satisfies $Q(0, 0) = Q(2s, 0) = 0$ pointwise. Thus a CF primitive exists. Moreover,

$$H^5(D_\infty, U(1)_s) = H^5(C_2, U(1)_s) \oplus H^5(C_2, U(1)_s) = 0, \quad (33)$$

because multiplication by two on $U(1)$ is surjective. The full phase therefore has a primitive as well. Pullback proves $2H \subset S$. An independent infinite-dihedral resolution constructs this complete tower and verifies its defining equations with corrected bar primitives.

Consequently the primitive free lattice is determined by the joint parity quotient $H/2H$, retaining the possible torsion shift of each free character. Let V be the projection of its surviving classes onto the free quotient modulo two. Choose a reduced row-echelon basis v_i of V and directly construct the flat tower of each row with its surviving torsion shift. Adjoin $2e_j$ in the nonpivot columns, using the universal doubled tower. These rows form an integral basis of the full surviving lattice, of index $2^{r-\dim V}$ in the original rank- r lattice. This construction requires no general integer-layer binary product. Earlier files without these witnesses retain only an abstract free splitting.

The accepted v09 archive **results/space_groups** contains complete classification and stacking results for all 230 groups. All 44 groups with nonzero free $p + ip$ rank export actual primitive lattice

bases and complete generator towers; all 32 surviving torsion $p + ip$ cases retain their full upper witnesses. The archive manifest records the frozen source and **normalized-pip-aw-edge-transport-v2** convention. This configuration enables the closed-CF phase optimization and native mod-two contraction, while retaining the integral bar comparison map. Its mathematical outputs agree literally with the complete v07 baseline. Constructed witnesses and their saved-certificate audits do not imply a separate full comparison-support audit of every group; only cases explicitly recorded in the validation log carry that additional numerical check.

9 Examples distinguishing graded layers and extensions

The new supplied space-group PDF lists four graded layers, with no stacking extension column. Its 920 entries agree with the independent 230-group graded calculation. A separate historical manuscript table contains four stacking differences despite agreement of all four graded layers:

Space group	Historical extension	Independent extension
68	$\mathbb{Z}_2^3 \oplus \mathbb{Z}_4$	$\mathbb{Z}_2^5$
81	$\mathbb{Z} \oplus \mathbb{Z}_2^4 \oplus \mathbb{Z}_8^3$	$\mathbb{Z} \oplus \mathbb{Z}_2^2 \oplus \mathbb{Z}_4 \oplus \mathbb{Z}_8^3$
82	$\mathbb{Z} \oplus \mathbb{Z}_2^4 \oplus \mathbb{Z}_8^2$	$\mathbb{Z} \oplus \mathbb{Z}_2^2 \oplus \mathbb{Z}_4 \oplus \mathbb{Z}_8^2$
101	$\mathbb{Z}_2 \oplus \mathbb{Z}_4$	$\mathbb{Z}_2^3$

For groups 68 and 101 there is one Majorana generator B and a bosonic kernel $\mathbb{Z}_2^k$, with $k = 4, 2$. The complete cochain reduction gives $2B = 0$, including nonzero incoming gauges and their phase primitives. The raw square need not vanish pointwise. A nonzero $2B$ would give exactly the historical alternative $\mathbb{Z}_4 \oplus \mathbb{Z}_2^{k-1}$. Because $2D = 0$ for each boson D , changing the lift $B \mapsto B + D$ cannot account for the difference.

For groups 81 and 82 the measured marked relation is

$$2P = C_1, \quad 2C_1 = 0, \quad (34)$$

with C_1 absent from all other lower relations. It gives a direct $\mathbb{Z}_4$ summand. There is an independent geometric consistency check: each actual affine group retracts onto its C_4 point group. The quotient and an order-four affine section were verified exactly. Consequently pullback and restriction force the C_4 classification to be a direct summand of the space-group answer.

On C_4 , write $s(a) = a \bmod 2$, $\eta(a) = \lfloor a/2 \rfloor$ and $B(a, b) = s(a)\eta(b)$. Then $dB = s^3$, and the CF part of the square is

$$K = B \cup_1 B + B \cup_2 dB + s \cup B, \quad K(a, b, c) = (a \bmod 2) \left\lfloor \frac{b+c}{4} \right\rfloor \bmod 2. \quad (35)$$

It evaluates to one on the mod-two bar cycle $\sum_{i=0}^3 [1|i|1]$, while s^3 evaluates to zero. This proves the nonzero CF square independently of the upper-phase choice. The complete finite calculation gives $\mathbb{Z}_4$, also listed in the supplied $\omega_2 = 0$ point-group table. The historical alternatives $\mathbb{Z} \oplus \mathbb{Z}_2^4 \oplus \mathbb{Z}_8^k$ contain no direct $\mathbb{Z}_4$ summand. These checks isolate an extension discrepancy; they do not diagnose which operation produced the historical value.

10 Verification and scope

The independent Python and GAP evaluators are checked against exact local lower towers and the supplied public formula evaluator. Separate checks use actual bar primitives from disputed space

groups. Local arithmetic agreement, universal coefficient identities, and physical normalization are recorded as different kinds of evidence.

The supplied normalized note proves its physical uniqueness statement relative to a fixed lower tower and generator, with explicit finite-group hypotheses. Using its cochains on a space group does not by itself extend the stated theorem. Any additional topological identification required for the crystalline application is to be stated separately from the arithmetic calculation.